

BARRAGE ANGLIERS, QC, Canada

WMO: 718201

Lat: 47.550N

Lon: 79.240W

Elev: 876

StdP: 14.24

Time Zone: -5.00 (NAE)

Period: 96-07

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.1	-16.6	-29.3	1.1	-21.0	-23.6	1.5	-16.1	18.8	10.1	17.3	11.3	3.6	90	0.589

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	85.3	68.5	82.2	66.5	79.0	65.0	71.3	81.6	69.3	78.3	67.5	75.2	8.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.0	106.2	75.4	66.2	99.6	73.7	64.5	94.0	72.5	35.7	81.2	34.0	78.6	32.5	75.2	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	16.3	14.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.5	7.3	11.6	21.4	37.4	51.9	62.8	65.8	63.8	57.2	43.4	30.6	18.3
	DBStd	22.60	13.76	12.46	12.65	9.24	8.93	8.18	6.07	6.20	7.47	8.61	8.66	11.99
	HDD50	5593	1324	1075	885	387	88	7	0	0	18	244	583	982
	HDD65	9575	1789	1495	1350	828	413	137	63	98	254	669	1032	1447
	CDD50	1744	0	0	0	9	148	392	490	427	236	41	1	0
	CDD65	250	0	0	0	0	8	71	88	61	21	1	0	0
	CDH74	2068	0	0	0	2	109	628	738	439	149	3	0	0
	CDH80	523	0	0	0	0	26	169	194	105	29	0	0	0
	WSAvg	6.7	6.6	6.6	7.1	7.2	7.2	6.2	6.0	6.0	6.6	7.3	7.4	6.7
	PrecAvg	38.0	2.5	2.0	2.4	2.5	3.1	3.9	4.0	4.2	4.0	3.3	3.1	2.9
(15) Precipitation	PrecMax	46.6	4.6	3.7	5.0	4.3	6.0	8.3	7.8	9.2	8.2	6.1	6.3	4.6
	PrecMin	29.6	0.6	0.6	0.4	0.6	1.4	1.5	1.1	1.5	1.6	1.1	0.4	1.2
	PrecStd	4.5	0.8	0.9	1.1	1.1	1.3	1.6	1.6	1.9	1.5	1.3	1.3	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.0	43.9	55.0	70.6	84.2	88.0	89.7	87.6	83.6	73.7	54.3	43.6
		MCWB	40.3	36.4	41.6	54.6	64.9	70.1	72.1	70.1	67.7	62.5	51.0	39.1
	2%	DB	36.0	37.5	48.8	63.8	76.5	84.9	84.7	82.9	78.1	66.8	51.5	39.4
		MCWB	34.1	32.9	41.8	48.3	57.2	68.0	68.2	68.2	66.1	59.2	48.9	37.0
	5%	DB	32.5	34.6	44.4	57.5	72.6	81.4	81.4	78.9	73.7	60.6	47.2	36.4
		MCWB	31.4	31.8	39.4	45.5	56.4	65.3	66.3	65.9	63.8	55.3	45.0	34.5
	10%	DB	28.6	31.7	40.1	52.0	67.6	77.4	78.1	75.4	69.8	56.2	42.3	33.8
		MCWB	27.4	29.9	35.1	43.1	55.3	64.0	64.8	64.0	61.6	51.9	39.7	32.3
	0.4%	WB	41.2	36.1	48.0	57.3	66.4	72.1	74.1	73.5	69.6	64.8	52.9	41.8
		MCDB	41.9	41.8	49.3	68.0	76.3	84.8	83.0	82.3	79.1	70.6	53.9	42.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.5	33.6	43.0	50.8	62.2	70.1	71.7	70.5	67.5	60.1	48.6	37.1
		MCDB	35.6	36.7	46.5	58.6	72.1	80.8	81.6	78.9	75.5	66.0	50.6	38.8
	5%	WB	31.4	32.0	39.4	46.7	58.9	68.2	69.2	68.3	65.6	56.0	44.8	34.4
		MCDB	32.6	34.1	44.4	56.9	68.2	77.3	78.0	75.3	72.3	60.2	47.1	36.1
	10%	WB	27.3	29.7	35.5	43.0	56.0	66.1	66.9	66.1	62.7	52.1	39.9	32.5
		MCDB	28.5	31.9	39.4	50.6	65.0	74.3	74.1	73.3	68.4	55.8	42.1	34.0
(35) Mean Daily Temperature Range	5% DB	MDBR	18.1	21.2	21.8	19.0	19.5	18.8	17.9	17.0	16.6	12.8	10.6	14.8
		MCDBR	17.8	19.9	22.4	29.4	26.9	24.7	22.8	22.1	22.0	20.5	15.9	12.5
		MCWBR	17.1	16.6	15.8	16.8	12.2	11.0	9.5	9.6	11.6	14.0	14.4	11.7
	5% WB	MCDBR	17.4	18.0	19.3	27.0	22.3	20.7	20.0	18.0	19.4	19.7	15.7	12.5
		MCWBR	17.2	15.4	14.8	16.7	12.6	10.7	9.9	9.2	11.1	14.4	14.7	11.9
(40) Clear-Sky Solar Irradiance	taub		0.265	0.284	0.315	0.355	0.361	0.389	0.395	0.384	0.364	0.329	0.303	0.264
	taud		2.276	2.215	2.237	2.323	2.395	2.343	2.354	2.378	2.413	2.497	2.450	2.350
	Ebn at Noon		259	278	286	285	286	278	274	272	267	261	245	246
	Edh at Noon		24	32	36	37	36	38	37	35	31	24	21	20
(44) All-Sky Solar Radiation	RadAvg		388	718	1194	1483	1656	1788	1754	1474	1116	663	346	277
	RadStd		58	107	129	125	141	121	145	102	112	88	38	28

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page